1. Write a query that uses a subquery to obtain all orders for the customer name Cisneros. Assume you do not know his customer number (cnum).

```
ERROR 1054 (42S22): Unknown column 'o.snum' in 'where clause'
D6_96715_bhagyesh>SELECT * FROM orders o WHERE o.snum=(SELECT snum FROM customers c WHERE cname ='Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)
```

7.
Extract all the orders of Motika.

```
D6_96715_bhagyesh>SELECT * FROM orders o WHERE o.snum =(SELECT s.snum FROM salespeople s WHERE s.sname ='Motik
a');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.06 sec)

D6_96715_bhagyesh>
```

8.
Find all the orders of salespeople servicing customers in London.

```
D6_96715_bhagyesh>SELECT * FROM orders o WHERE o.snum = ANY(SELECT  c.snum FROM customers c WHERE c.city='Lond
on');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.02 sec)
```

6.
Write a query that selects all orders for amounts greater than any for the customers in London.

```
D6_96715_bhagyesh>SELECT * FROM orders WHERE amt > Any(SELECT amt FROM orders  WHER
E cnum in (SELECT cnum FROM customers  where city ='London'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.00 sec)
```

4.

Write a query that selects all customers whose ratings are equal to or greater than ANY
of Serres'. (Hint: find all customers of serres salesperson).

```
D6_96715_bhagyesh> SELECT * FROM customers WHERE rating >= ANY (SELECT rating FROM
customers WHERE snum =(SELECT snum FROM salespeople WHERE sname='Serres'));
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
```

11.
Select customers who have a greater rating than any other customer in Rome.

```
D6_96715_bhagyesh>SELECT * FROM customers WHERE rating > ANY (SELECT rating FROM cu
stomers WHERE city='Rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
```

12.
Select all orders that had amounts that were greater that at least one of the orders from
'1990-10-04'.

```
D6_96715_bhagyesh>SELECT * FROM orders WHERE amt > ANY(SELECT amt FROM orders WHERE
 odate = '1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
```

13.
Find all orders with amounts smaller than any amount for a customer in San Jose.

```
D6_96715_bhagyesh>SELECT * FROM orders WHERE amt < ANY(SELECT amt FROM orders WHERE
 cnum =ANY (SELECT cnum FROM customers WHERE city ='San Jose'));
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.01 sec)
```

2.
Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
D6_96715_bhagyesh>SELECT o.amt , c.cname, c.rating FROM customers c INNER JOIN orde
rs o ON c.cnum = o.cnum  where o.amt > (SELECT AVG(amt) FROM orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.00 sec)
```

14.
Select those customers whose rating are higher than every customer in London.

```
D6_96715_bhagyesh>SELECT * FROM customers WHERE rating > ALL(SELECT rating FROM cus
tomers  WHERE city ='London');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

5.
Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
D6_96715_bhagyesh>SELECT sname,snum,city FROM salespeople s WHERE city NOT IN (SELE
CT city FROM customers where snum=s.snum);
+---------+------+-----------+
| sname   | snum | city      |
+---------+------+-----------+
| Motika  | 1004 | London    |
| Rifkin  | 1007 | Barcelona |
| Axelrod | 1003 | New York  |
+---------+------+-----------+
3 rows in set (0.00 sec)
```

9.
Find names and numbers of all salesperson who have more than one customer.

```
D6_96715_bhagyesh>SELECT sname,snum FROM salespeople WHERE snum IN (SELECT snum FRO
M customers GROUP BY snum HAVING count(snum) > 1);
+--------+------+
| sname  | snum |
+--------+------+
| Peel   | 1001 |
| Serres | 1002 |
+--------+------+
```

10.
Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
D6_96715_bhagyesh>SELECT s.snum, s.sname, s.city, COUNT(c.cnum) c FROM salespeople s INNE
R JOIN customers c ON s.snum = c.snum GROUP BY s.snum, s.sname, s.city HAVING COUNT(c.cnu
m) > 1;
+------+--------+----------+---+
| snum | sname  | city     | c |
+------+--------+----------+---+
| 1001 | Peel   | London   | 2 |
| 1002 | Serres | San Jose | 2 |
+------+--------+----------+---+
```

3.
Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
D6_96715_bhagyesh>SELECT snum, SUM(amt) as s_amt FROM orders GROUP BY snum HAVING SUM(amt
) > (SELECT MAX(amt) FROM orders);
+------+----------+
| snum | s_amt    |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.04 sec)
```